Marketing Measurement

From Attribution to Influence — Lecture Handout

Universidad Galileo student conversation • October 28, 2026 • Mohammed Jawhar

Attribution is not incrementality

Attribution assigns credit for a conversion to marketing touchpoints. Incrementality asks the counterfactual question: what would have happened without a particular intervention? The two questions are related but not interchangeable.

Example: a buyer watches YouTube, sees a Meta reminder, then clicks Search to buy. Last-click attribution may credit Search even though earlier activities—and baseline interest—also influenced the decision.

What Marketing Mix Modeling (MMM) contributes

MMM uses aggregate time-series (and optionally geographic) variation to estimate contributions from media, baseline demand and carefully selected business conditions. It can support channel comparisons, spend-response curves, and budget scenarios.

Business factors: seasonality, promotion, prices, consumer demand and competitor activity. The central modeling skill is choosing controls that address relevant confounding—not including every available metric or adjusting for variables caused by advertising.

The Google Meridian classroom example

Data	156 weeks of synthetic US e-commerce data (2023–2025)
Channels	Google Search, Meta, YouTube
Outcome	Revenue in USD (not a conversion-count outcome)
Other inputs	Impressions, spend, demand index, promotion flag
Outputs	Model fit, R-hat, contributions, response curves, budget scenarios

Teaching rule: if you increase budget, estimated incremental revenue may increase while additional revenue per new dollar decreases. For a clean reallocation example, hold total spend fixed.

Frequentist vs. Bayesian: quick comparison

Question	Frequentist	Bayesian
Unknown parameters	Fixed unknown coefficients	Posterior distributions
Uncertainty	Confidence intervals	Credible intervals

Advantage	Efficiency and long-run error properties	Priors and direct probability statements
Caution	Not assumption-free	Not automatically causal

Coin illustration: 7 heads in 10 tosses. The frequentist point estimate of heads probability is 0.70. With a uniform Beta(1,1) prior, the Bayesian posterior is Beta(8,4), implying about an 88.7% probability the coin favors heads (given the prior and binomial model).

How should we read Meridian diagnostics?

An R-hat near 1 suggests different MCMC sampling chains agree. A high R^2 or low MAPE suggests good descriptive fit. Neither alone verifies causal attribution. Check priors, credible intervals, effective sample sizes, divergences, calibration, and model assumptions.

From the 4Ps to the 7Ps to the proposed 7Gs

4Ps: Product, Price, Place, Promotion. 7Ps adds People, Process, Physical Evidence. A recent research paper proposes 7Gs to discuss AI-generated offerings, value calibration, access, persuasion, agency networks, experiences and trust signals.

This proposed 7G marketing-strategy framework is distinct from Meridian's statistical Marketing Mix Modeling. The paper raises conceptual questions rather than establishing that AI improves marketing results.

The closing question

If generative AI can influence what customers see and buy, how do we measure genuine incremental value—while preserving accuracy, privacy, fairness and trust?

Open the full materials

Slides, Jupyter notebook, synthetic dataset, summaries and reading notes are included in this project.
Research paper:
https://www.researchgate.net/publication/399671440_Marketing_mix_rebuilding_for_the_age_of_generative_AI_From_7P_to_7G_marketing_mix_model

Reminder: the dataset is synthetic, and the notebook's posterior calculations should be run and validated in a compatible Meridian environment before presenting them as reproducible results.